Attention-Deficit Hyperactivity Disorder

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Introduction

Attention-deficit hyperactivity disorder (ADHD) is the most common pediatric neuropsychiatric disorder; it affects 4–12% of North American school-aged children[1] and 2–8% of preschool aged children (3–5 years of age).[2] It is characterized by 3 hallmark symptoms—inattention, hyperactivity and impulsivity—present at a magnitude and frequency beyond what would be expected for a given age or developmental stage.[1][3] Symptoms often interfere with normal development; are present in 2 or more settings, such as at home, work or school; and have a direct, unfavourable impact on social, academic, cognitive or occupational functioning.[4] Individuals may be diagnosed with the primarily inattentive subtype, the primarily hyperactive/impulsive subtype or a combined subtype, which is most common.[4][5] See Table 1 for a list of the different ADHD symptoms and diagnostic criteria.

Symptoms of ADHD typically present before the age of 12 and may persist into adulthood.[3][4] Although the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision* (DSM-5-TR) discusses symptom onset before 12 years of age only, there is emerging evidence that ADHD may occur in young adults with no childhood diagnosis.[6][7] The prevalence of ADHD in adulthood is approximately 3–4%.[4]

People with ADHD are at higher risk than the general population for comorbid psychiatric disorders, such as oppositional defiant disorder (ODD); learning disorders; and mood, anxiety and substance use disorders.[4][5][8][9]

Table 1: *DSM-5-TR Criteria for Diagnosing ADHD*

A.		A persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning or development, as characterized by (1) and/or (2):
	1.	*Inattention*: Six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent with developmental level and that negatively impacts directly on social and academic/occupational activities.[a] For older adolescents and adults (age 17 and older), at least five symptoms are required. a. Often fails to give close attention to details or makes careless mistakes in schoolwork, at work or during other activities (e.g., overlooks or misses details, work is inaccurate) b. Often has difficulty sustaining attention in tasks or play activities (e.g., has difficulty remaining focused during lectures, conversations or lengthy reading) c. Often does not seem to listen when spoken to directly (e.g., mind seems elsewhere, even in the absence of any obvious distraction) d. Often does not follow through on instructions and fails to finish schoolwork, chores or duties in the workplace (e.g., starts tasks but quickly loses focus and is easily sidetracked) e. Often has difficulty organizing tasks and activities (e.g., difficulty managing sequential tasks; difficulty keeping materials and belongings in order; messy, disorganized work; has poor time management; fails to meet deadlines) f. Often avoids, dislikes or is reluctant to engage in tasks that require sustained mental effort (e.g., schoolwork or homework; for older adolescents and adults, preparing reports, completing forms, reviewing lengthy papers) g. Often loses things necessary for tasks or activities (e.g., school materials, pencils, books, tools, wallets, keys, paperwork, eyeglasses, mobile telephones) h. Is often easily distracted by extraneous stimuli (for older adolescents and adults, may include unrelated thoughts) i. Is often forgetful in daily activities (e.g., in doing chores, running errands; for older adolescents and adults, returning calls, paying bills, keeping appointments)
	2.	*Hyperactivity-impulsivity*: Six (or more) of the following symptoms have persisted for at least 6 months to a degree that is inconsistent with developmental level and that negatively impacts directly on social and academic/occupational activities.[a] For older adolescents and adults (age 17 and older), at least five symptoms are required. a. Often fidgets or taps with hands or feet or squirms in seat b. Often leaves seat in situations when remaining seated is expected (e.g., leaves his or her place in the classroom, in the office or other workplace, or in other situations that require remaining in place) c. Often runs about or climbs in situations where it is inappropriate. (Note: in adolescents or adults, may be limited to feeling restless) d. Often unable to play or engage in leisure activities quietly e. Is often "on the go" acting as if "driven by a motor" (e.g., is unable to be or uncomfortable being still for extended time, as in restaurants, meetings; may be experienced by others as being restless or difficult to keep up with) f. Often talks excessively g. Often blurts out an answer before a question has been completed (e.g., completes other people's sentences; cannot wait for turn in conversation) h. Often has difficulty waiting for his or her turn (e.g., while waiting in line)

		i. Often interrupts or intrudes on others (e.g., butts into conversations, games or activities; may start using other people's things without asking or receiving permission; for adolescents and adults, may intrude into or take over what others are doing)
	B.	Several inattentive or hyperactive-impulsive symptoms were present prior to age 12 years.
	C.	Several inattentive or hyperactive-impulsive symptoms are present in two or more settings (e.g., at home, school or work; with friends or relatives; in other activities).
	D.	There is clear evidence that the symptoms interfere with, or reduce the quality of, social, academic or occupational functioning.
	E.	The symptoms do not occur exclusively during the course of schizophrenia or another psychotic disorder and are not better explained by another mental disorder (e.g., mood disorder, anxiety disorder, dissociative disorder, personality disorder, substance intoxication or withdrawal).

[a] Note: the symptoms are not solely a manifestation of oppositional behaviour, defiance, hostility or failure to understand tasks of instructions.

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Goals of Therapy

- Eliminate or significantly decrease ADHD symptoms
- Improve behavioural, academic and/or occupational performance
- Improve self-esteem and social functioning
- Prevent or minimize the impact of potential complications (e.g., comorbid mood, anxiety and/or substance use disorders)
- Minimize adverse effects of medications
- Improve quality of life

Investigations

- Currently there are no objective tests that diagnose ADHD.
- The diagnosis of ADHD is based on evidence of 6 or more specific behaviours/symptoms (see Table 1) for children and at least 5 in those over 17 years of age. These symptoms will have persisted for at least 6 months to a degree that is inconsistent with developmental level and impact overall functioning.[4]
- Gather information from the patient, family/caregivers, teachers/assistants and psychologists (where applicable) to identify:
  - nature and duration of ADHD symptoms
  - impact of symptoms on functioning
  - age of onset
  - social interactions with family and peers
  - situations that exacerbate or ameliorate symptoms
  - symptoms suggestive of other potentially comorbid disorders, such as ODD, conduct disorder, mood/anxiety disorders, tic disorders and/or learning disabilities. If present, consider referring to a specialist and/or subspecialist for diagnosis of complex ADHD
  - assessment for substance use disorders; it may be indicated in some patients to obtain urine drug screens both prior to initiating treatment with stimulant and over the course of treatment
- Various assessment/rating scales can be used to collect the above information and assess progress; see the Canadian ADHD Resource Alliance (CADDRA) Assessment Toolkit to find the most appropriate assessment vehicles and treatment suggestions for individual patients.[8] The toolkit includes the SNAP-IV Teacher and Parent Rating Scale, which is particularly useful for parents and/or teachers.
- Birth/prenatal history including premature birth and in utero exposure to substances.[10]
- Physical examination/medical history to assess hearing, vision, thyroid function, neurologic status, cardiac function, history of dysmorphic disorder (such as Down syndrome) or anemia.
- Prior to initiating stimulants, screen for cardiovascular risk factors via a detailed medical history:[11]
  - baseline blood pressure, heart rate, height/weight; re-assess at follow-up
  - routine ECG monitoring, either before or after starting medication, is not recommended in young patients without history of heart disease and with normal physical examination;[8] however, it is reasonable to consider in some patients, such as those with concurrent cardiac abnormalities (including raised BP, abnormal heart rate) or those with a family history of cardiac disease or sudden death
  - cardiology consultation recommended in patients with established or suspected heart disease

Therapeutic Choices

An algorithm for the management of ADHD is presented in Figure 1.

A multimodal treatment approach of combining nonpharmacologic (e.g., behaviour modification programs) and pharmacologic therapies constitutes the best treatment strategy for children and adults with ADHD.[12][13][14][15] However, management of behaviour by way of parent or teacher training alone is considered first line in preschool-aged children.[16][17] Addition of methylphenidate is reserved for preschoolers with moderate to severe ADHD who do not respond to behavioural therapy if the benefit outweighs the harm; few studies have examined the long-term effects of stimulant use in preschool-aged children.[18] Developing a shared-care approach that incorporates the patient's and caregivers' values and preferences into goals of therapy and treatment approaches is important, considering that ADHD is chronic and often present with psychiatric and developmental comorbidities.[12]

Nonpharmacologic Choices

Some clinicians, parents and/or patients may prefer, and require guidance about, nonpharmacologic approaches such as behaviour management, cognitive-behavioural therapy, parent training, organizational skills training, etc. The CADDRA 2020 guidelines recommend incorporating these modalities into the patient treatment plan, and provide extensive advice and practical interventions for the clinicians and family members.[8] Evidence for the use of nonpharmacologic therapies alone is lacking compared to the higher-quality RCTs conducted on stimulants.[19] Compared with stimulants, nonpharmacologic options appear to be less effective at reducing the core symptoms of ADHD.[5][14][20] Combined behavioural/pharmacologic therapy was shown to be more effective at reducing oppositional behaviours and anxiety while improving social interactions and self-esteem when compared to either treatment strategy alone.[14]

- Behavioural therapies are designed to minimize negative behaviours and promote positive ones by teaching parents (and sometimes teachers) techniques to improve a child's behaviour.[3]
    - Behavioural therapies play an important role in improving social and family interactions, self-esteem and the common behaviours seen in ADHD.[14][20] Behavioural strategies that included sleep hygiene practices reduced the severity of parent- and teacher-reported ADHD symptoms and sleep problems at 3 and 6 months postintervention.[21]
- Social skills, mindfulness and parent-management training may all be offered as formalized training interventions, depending on the patient and family needs.[8] Parental training may increase confidence and reduce stress for parents.[22]
- A meta-analysis of exercise interventions (e.g., short-term aerobic exercise and yoga) demonstrated an improvement in core ADHD symptoms, a reduction in anxiety and improved cognitive functioning.[23]
- The elimination of certain foods from the diet in an effort to minimize ADHD symptoms is based mainly on observations that some children exhibit more severe hyperactive behaviours in association with certain sugars, dyes or preservatives.[24][25] Evidence for diets eliminating these foods or additives is not robust enough to make any specific dietary recommendations.
- EndeavorRx ADHD video game is a novel treatment modality that rewards appropriate decision-making in patients with ADHD.[26] It was studied in over 600 children with ADHD and is FDA-approved in those 8–12 years of age as adjunctive therapy for ADHD.[26] EndeavorRx is not available in Canada; however, EndeavorOTC is a similar option available for adults with ADHD.[27]

Pharmacologic Choices

Reserve medication for patients with a clear diagnosis of ADHD and who have demonstrated impairment in learning or academic/occupational performance due to their attention difficulties, or whose behaviours and social interactions are impaired by a lack of impulse control and/or hyperactivity. Medications used in the management of patients with ADHD are listed in Table 3 and Table 4. Stimulant medications are considered the most effective treatment for the core symptoms of ADHD (hyperactivity, impulsiveness and inattention) and are usual first-line agents.[8][14][28]

Stimulants

Long-acting stimulant medications (**lisdexamfetamine**, **methylphenidate** controlled-release formulations, **mixed salts amphetamine**) are first-line agents for treating ADHD.[3][5][8] They are controlled substances and their dispensing is subject to specific regulations. Their efficacy at reducing core ADHD symptoms has been demonstrated in a wide range of patients from 6 years of age to adult.[28][29][30] Controlled trials consistently demonstrate that at least 70% of patients receiving stimulant therapy will have a clinically significant decrease in core ADHD symptoms.[3][5][14] Limited data from short-term studies suggests that stimulants improve parent-reported quality of life and academic achievement and lower rates of comorbid anxiety and depression in young adulthood;[31][32][33][34] stimulants have not been adequately studied in terms of important long-term outcomes such as quality of life, employment, school completion and long-term morbidity or mortality.[12]

Stimulants are available in immediate-, intermediate- and extended-release formulations. There are minimal differences in efficacy and safety among stimulants. A meta-analysis comparing stimulants supported amphetamines as preferable first-line agents for adults and methylphenidate for children; however, these conclusions were based on limited evidence.[35] Therefore, the choice of stimulant depends on patient and physician preferences.[12]The CADDRA 2020 guidelines do not recommend any specific stimulant as first-line therapy; however, they do recommend long-acting stimulants as preferred first-line agents.[8] Immediate- and intermediate-acting stimulants are recommended for use when the patient requires more flexible dosing and minimal hours/day of medication or as add-on therapy to an extended-release formulation.

A 3- to 4-week trial with stimulants is reasonable, although improvement of core ADHD symptoms is often observed in the first week of therapy. If a patient responds adequately with minimal adverse effects, it is reasonable to continue therapy for 6–12 months, depending on patient circumstances. Patients who do not tolerate or respond adequately to the optimal dose of the initial stimulant after 4 weeks of therapy should be switched to an alternative stimulant. For school-aged children and adolescents, consider switching during holidays or after a report card period, to minimize disruption and aid in evaluating the effect of the medication change.[8] There are no general guidelines on switching from one stimulant to another; several factors require consideration, including duration of action and other pharmacokinetic properties, salt formulations and patient age/weight.[36] Consult individual product monographs for guidance; re-titration of the new agent is often required.

Contraindications to stimulants include history of hypersensitivity to sympathomimetic amines, symptomatic cardiovascular disease (including moderate to severe hypertension, advanced atherosclerosis), uncontrolled hyperthyroidism, history of drug abuse and concurrent use with an MAOI.

Modafinil is a CNS stimulant that promotes wakefulness in the treatment of narcolepsy but it is not approved for use in ADHD. While there is some evidence from double-blind trials in children and adults that modafinil (170–425 mg/day) is superior to placebo in decreasing core symptoms of ADHD,[37][38] it is less effective than other stimulants in ADHD and is not often used.

Long-Acting Stimulants

The CADDRA 2020 guidelines recommend long-acting stimulants as first-line therapy for ADHD.[8] Long-acting formulations of **mixed salts amphetamine** (Adderall XR), **methylphenidate** (Biphentin, Concerta, Foquest, Quillivant ER and Jornay PM) and **lisdexamfetamine** (Vyvanse) have a duration of action of 8–14 hours and are as effective as appropriately dosed shorter-acting stimulants.[28] Advantages of these long-acting products include single daily dosing, potential for improved adherence, avoidance of the need for medication administration at school and decreased risk of rebound hyperactivity.[39] Delayed-release methylphenidate (Jornay PM) is the only formulation administered in the evening; it contains delayed-release microbeads with an onset of action the following morning and effects that last most of the day.

Intermediate-release stimulants include methylphenidate sustained-release tablets, e.g., Ritalin SR and **dextroamphetamine** sustained-release spansules (Dexedrine Spansules). They have a longer duration of action (up to 8 hours) than their immediate-release counterparts, but often require more than once-daily dosing. A small portion of patients may require additional afternoon doses of immediate-release stimulants to help with homework and to control early-evening symptoms. Intermediate and immediate-release stimulants are recommended as second-line treatment options.[8]

Adverse Effects of Stimulants

The most common adverse effects of stimulants are increased heart rate and blood pressure, GI upset, appetite suppression, anxiety, irritability and insomnia.[8] For more information, see Table 2.

A 3-year follow-up of the participants of the MTA study found that children given stimulants were, on average, 2 cm shorter and weighed 2.7 kg less than children who were never medicated during the same 3 years.[40] However, reassuring results from a 2023 study following 1410 patients demonstrated no relationship between methylphenidate use and growth suppression.[41] Nevertheless, monitor children taking stimulants for potential growth suppression—record weight and height at baseline and then every 3–6 months. Plot on growth charts and compare measurements with established norms for a given child's age/gender.

When monitoring efficacy and adverse effects of ADHD medications, use standardized checklists and/or rating scales and collect the information from 2 or more settings (e.g., school and home).[8][12]

Patients with ADHD may have a slightly increased risk of suicidal thoughts and behaviours. Reports of suicide-related events in patients taking stimulants and atomoxetine have prompted warnings that the medications may contribute to this risk in some patients.[42] The reported events occurred at varying times during treatment, but the greatest concern was at the start or end of treatment, or when the dosage was adjusted. Patients, family and friends should watch for these symptoms and report them promptly. Health-care professionals should monitor patients on ADHD medications for suicidal thoughts and consider a change in treatment if concerns arise.

Both methylphenidate and amphetamines have been associated with increased risk of psychosis in patients with ADHD. An observational study suggested the risk of new-onset psychosis in adolescents and young adults with ADHD taking stimulants was 1 in 660, and that the rate was higher in those taking amphetamines compared to methylphenidate.[43][44] As a precaution, consider nonstimulant options (e.g., atomoxetine and/or guanfacine) first in patients with a history of psychosis.

Health Canada and regulatory agencies in other countries have issued advisories about the possible association of stimulants (and atomoxetine) with cardiovascular risks, such as sudden cardiac death in those with underlying cardiac anomalies, and with adverse psychiatric symptoms, such as hallucinations and agitation. The FDA and Health Canada have issued warnings that stimulants and atomoxetine may be associated with rare reports of priapism in male children, adolescents and adults taking these medications.[45][46]

Screen patients for risk of sudden cardiac death before prescribing stimulants (see Investigations); if risk factors are present, referral to cardiology is warranted.[11] Inform patients (and parents/caregivers where appropriate) about potential adverse effects of stimulants on sleep, appetite, cardiovascular or psychiatric health, the possibility of prolonged erections (>2 h), and the need to seek medical attention if these events occur. The CADDRA ADHD Assessment Toolkit contains useful tools for monitoring adverse effects.[8]

Diversion and Substance Abuse

Some clinicians have concerns about the risk of substance abuse in patients with ADHD who are treated with stimulants. A diagnosis of ADHD is a risk factor for substance-use disorder and smoking dependence.[47] One meta-analysis indicated that children with ADHD who were treated with stimulants had a lower risk of substance-use disorders (drug and alcohol) later in life when compared with those who were untreated.[48] Similarly, a meta-analysis in adolescents and adults with ADHD suggested that treatment with stimulants or atomoxetine was associated with lower concurrent and long-term risks of substance-related problems.[49] Another meta-analysis reported an association between stimulant therapy and lower rates of cigarette smoking.[50] If diversion of stimulant medication is suspected, consider switching to an agent with less abuse potential (e.g., atomoxetine, guanfacine).[51] When prescribing stimulants, educate patients with ADHD about the dangers associated with misuse and diversion of these medications.[52]

When monitoring patients taking stimulants for diversion and substance abuse, health-care providers should consider the following:[53][54][55]

- History of substance use
- Other comorbid conditions associated with substance abuse (e.g., depression, anxiety, bipolar disorder)
- Regular follow-up to assess for response to treatment
- Behaviours associated with substance abuse (dilated pupils, insomnia, increased irritability, weight loss, changes in social interactions, increased requests for refills or dosage increases)
- Collaboration with addictions specialists if concerns arise

Atomoxetine

Atomoxetine, a norepinephrine reuptake inhibitor, is recommended as a second-line agent for the treatment of children ≥6 years of age, adolescents and adults with ADHD.[8] It is not classified as a stimulant and is not a controlled substance. The efficacy and tolerability of atomoxetine have been studied in several well-designed trials.[56][57][58][59][60][61][62] RCTs confirm that after 6–12 weeks of treatment, atomoxetine reduces core ADHD symptoms by at least 25–30% in 60–70% of individuals.[57][60][62] The efficacy of atomoxetine approaches that of stimulants, although it may take 3–4 weeks to see its beneficial effects. While some guidelines list it as a first-line option, the available evidence supports a role in therapy for those who have either not responded to or not tolerated an adequate trial of stimulant medications. It should also be considered for those with ADHD and comorbid substance-abuse disorder or anxiety.[28][36] Contraindications to atomoxetine include hypersensitivity to atomoxetine, narrow angle glaucoma, history of severe cardiac or vascular disorders, pheochromocytoma, and concurrent use with an MAOI.

Alpha$_2$-adrenergic Agonists

The alpha$_2$-adrenergic agonists **guanfacine** and **clonidine** are additional treatment options for ADHD.[3][28] They primarily reduce symptoms of aggression, impulsivity and hyperactivity and have less pronounced benefits on inattention. When used concurrently with stimulants, clonidine and guanfacine often target sleep disruptions, aggression, impulsivity, comorbid ODD and tics.[10][56]

Guanfacine (extended-release) is a selective alpha$_2$-adrenoceptor agonist officially indicated for the treatment of ADHD in children 6–17 years of age; it is recommended as a second-line agent for ADHD.[8] It can be used as monotherapy or as adjunctive therapy with stimulants. Compared with clonidine, guanfacine has more selective neuronal activity and a longer duration of action, resulting in less sedation and less hypotension. Several RCTs and open-label studies in children with ADHD have demonstrated that guanfacine 1–4 mg/day has a clinically and statistically superior effect on ADHD symptoms compared with placebo.[63][64][65] There are no adequate head-to-head comparisons of guanfacine with stimulants or the less-expensive clonidine. The most common adverse effects of guanfacine are sedation and headache. Data comparing guanfacine to other ADHD treatments are limited.

Clonidine is recommended as a third-line agent for ADHD.[8] Its use is supported by few studies; a meta-analysis on the clinical use of clonidine for ADHD suggests it has a moderate-size effect on symptoms of ADHD and tics, although this is not based on head-to-head trials.[66]

Antidepressants

In general, antidepressants are considered less effective than stimulants in the management of ADHD in children; hence, they are considered third-line options or as adjunctive therapy.[3][8][28] Antidepressants may benefit patients with comorbid conditions such as depression, anxiety, enuresis or tic disorders.[56]

Bupropion, a norepinephrine and dopamine reuptake inhibitor, is moderately effective for the treatment of ADHD in both children and adults.[56][67][68][69] Although there are insufficient data from RCTs to recommend its use, some evidence suggests **venlafaxine**, a serotonin-norepinephrine reuptake inhibitor, may be helpful in the management of ADHD, particularly in adults.[28][36][70]

Many controlled trials have examined the effects of tricyclic antidepressants (TCAs), such as **desipramine**, **imipramine** and **nortriptyline**, in the short-term treatment of ADHD.[71] TCAs are less effective than stimulants at reducing the core symptoms of ADHD. Despite the fact that most TCA studies have significant design flaws, such as open label design or lack of randomization, the benefits of TCAs may outweigh their risks in some patients who cannot take stimulants, atomoxetine or bupropion.[10][28][56][71]

Antipsychotics

A systematic review of the role of second-generation antipsychotics (SGAs), primarily **risperidone**, in ADHD failed to confirm that these agents benefit this patient population.[72] SGAs are sometimes used to decrease behaviours seen in children with comorbid conduct disorder, ODD, autistic disorders, impulse control disorders and Tourette syndrome. Antipsychotics are associated with several serious adverse effects, such as metabolic syndrome, extrapyramidal symptoms, and increased risk of unexpected mortality; some may also negatively affect cognition in patients with ADHD.[73] Thus, the benefits of antipsychotics for ADHD are often outweighed by their risks.

Natural Health Products

Although evidence from well-designed randomized trials is sparse, various natural health products, such as herbs, vitamins and nutritional supplements, have been tried on the basis of their traditional uses. For example, agents with possible anxiolytic, sedative or hypnotic effects, e.g., **chamomile**, **valerian**, **melatonin**, have been used in children who are restless, anxious or having sleep difficulties.[24] These herbs may play a role in calming a hyperactive child or promoting sleep in children with ADHD and insomnia. Other agents of interest include antioxidants such as **blue-green algae**, **omega-3 polyunsaturated fatty acids**, various **B vitamins**, **ginkgo biloba**, **pycnogenol** and **evening primrose oil** (essential fatty acids). In general, concerns regarding efficacy and quality of natural health products make it difficult to support their use in place of existing prescription medications; however, after evaluating potential drug interactions and clinical risks, it may be possible to use some of these products safely in conjunction with approved therapies.

Management of Common Adverse Effects of ADHD Treatment

Table 2: Management of Common Adverse Effects of ADHD Treatment

Adverse Effect	Monitoring	Management	Implicated Medications
Appetite suppression	Monitor for consistent appetite suppression and changes in weight Q 2 wk for the first 2 months, then Q 6 months. In children and adolescents, monitor height as well.	Administer medications during or after meals, not before. Maximize nutrition content when patient is not having symptoms of appetite suppression (e.g., evenings, before the morning dose of stimulant). Reduce portions and increase snacking times. Consider nutritional meal supplements. Consider using shorter acting stimulants to allow for return of appetite later in the day. Consider drug holidays on weekends or during vacations.	Stimulants, atomoxetine, bupropion.
Cardiovascular (increased HR, BP)	Monitor BP and HR in the first 2 wk of starting a stimulant medication, then Q 3 months. ECG not needed routinely if no previous history of cardiovascular disorder.	If significant changes occur in BP, HR or ECG, discontinue and consider consulting a cardiologist. These include sustained tachycardia, arrhythmia, or systolic blood pressure > 95th percentile.	Stimulants cause increased HR/BP. Alpha-2 agonists cause decreased HR/BP. They should always be tapered; if stopped abruptly, they may cause a hypertensive crisis. TCAs may cause tachycardia.

Adverse Effect	Monitoring	Management	Implicated Medications
Psychiatric (anxiety, irritability, insomnia, tics)	Monitor for difficulties falling asleep, staying asleep and/or early morning awakenings at 1 wk, then monthly for the first 3 months, then Q 6 months. Caregivers may use the Sleep Disturbance Scale for Children or the Children's Sleep Habits Questionnaire to monitor at home.	Often worse upon initiation and resolves after 1–2 wk of therapy. For insomnia, may need to lower the stimulant dose, change the time to an earlier administration, change to a shorter-acting formulation, add sedating medication at bedtime (e.g., melatonin, antihistamine) or discontinue the offending stimulant. Minimize use of caffeine and other psychostimulants. Limit stimulating activities (e.g., use of electronic devices) in the evenings.	Stimulants, atomoxetine, bupropion, venlafaxine.

Abbreviations

BP = blood pressure; ECG = electrocardiogram; HR = heart rate; TCA = tricyclic antidepressant

Drug Holidays

While ADHD is a lifelong condition for most individuals, symptoms may dissipate as patients enter adolescence. Weaning the medication for a 2- to 3-week period once a year (usually during work or school holidays) may provide an opportunity to reassess ADHD-related behaviours and to confirm whether the stimulant is still required for the next school term. Extended drug holidays, e.g., several months over the summer holidays, are generally not recommended in children with moderate to severe ADHD symptoms who are doing well on the medication. The return of symptoms and resultant effects on behaviour and self-esteem do not typically outweigh the risks of taking the medication. Drug holidays (at times of low environmental stresses) may be useful when adverse effects (such as growth suppression or weight loss >10% of initial body weight) have occurred or when attempting to assess the continued benefit.[75]

Abrupt discontinuation of stimulants may cause withdrawal symptoms in some individuals[8], especially if they have been treated for prolonged periods and/or at maximum doses; consider tapering over several weeks in patients who poorly tolerate discontinuation or have been on medication for longer than 3 months. Alpha$_2$-adrenergic agonists must be tapered slowly (i.e., 0.1 mg/wk for clonidine and 1 mg/wk for guanfacine) due to risk of rebound hypertension if abruptly discontinued. Atomoxetine is associated with less withdrawal than other agents.

Choices during Pregnancy and Breastfeeding

ADHD and Pregnancy

The impact of pregnancy on ADHD symptoms is not well defined. It is possible that hormonal variations related to pregnancy and psychological and emotional issues triggered by impending parenthood could worsen ADHD symptoms.[76] In addition, there is a paucity of data about the effects of stimulant medications on pregnancy and fetal outcomes.[77] This is primarily because data regarding exposure to stimulants during pregnancy and an assessment of the fetal and other outcomes, particularly specific congenital malformations, are rare.

Given that prescription medications to treat ADHD are being used in patients of childbearing age more frequently than a decade ago, it is important to discuss potential risks and benefits of treatment with patients who are pregnant or planning a pregnancy.

Management during Pregnancy

For patients with mild-moderate symptoms without functional impairments, nonpharmacologic treatment is preferred (see Nonpharmacologic Choices).

Given the limited available data regarding pharmacologic treatment of ADHD during pregnancy, clinicians must weigh the risks versus potential benefits of treatment for each patient.

An observational cohort study of 1.8 million pregnancies found that use of **amphetamines** (**amphetamine** or **dextroamphetamine**) during pregnancy was not associated with an increased risk of cardiac or major congenital malformations.[78] It also found that **methylphenidate** exposure was not associated with an overall increased risk of major congenital malformations, although it appeared to be associated with an elevated risk of cardiac malformations.[78] The clinical significance of these findings remains unclear, but caution is advised with the use of methylphenidate, especially during the first trimester. There are no reports of **lisdexamfetamine** use during pregnancy.

TCAs, **bupropion** and **venlafaxine** should be used with caution and appear to be less effective for ADHD symptoms than other medications.[76][79][80]

Until more data are available, **atomoxetine** should be avoided during pregnancy, at least during the first trimester.[79]

Clonidine has not been associated with birth defects, although there is little data about exposure during the first trimester.[79] There are no data on the use of **guanfacine** during pregnancy.

Because of a lack of evidence to assess the safety of **risperidone** during pregnancy[79] and its limited benefits in treating ADHD, it should be avoided for this indication in pregnant patients.

ADHD and Breastfeeding

As in pregnancy, the risks of not treating the patient during the postpartum period must be weighed against the potential risks associated with pharmacologic treatment; nonpharmacologic treatment is preferred when possible (see Nonpharmacologic Choices).

Transfer of **methylphenidate** into breast milk appears to be low and might not affect the nursing infant adversely, although the effects on neurological development have not been well studied.[81] The nursing infant should be monitored for agitation and poor weight gain.[76] **Amphetamines** (**amphetamine**, **dextroamphetamine**, **lisdexamfetamine**) are transferred into breast milk, and their impact on the infant is unknown.[81] Breastfed infants should be monitored for insomnia, agitation, irritability and poor weight gain.[76]

Maintaining treatment with **TCAs**, **bupropion** or **venlafaxine** during breastfeeding for patients with a good response to one of these agents can be considered.[76] Monitoring the nursing infant for sedation, weight gain and irritability is recommended.[76]

There are no reports of **atomoxetine** use during breastfeeding.[81]

Clonidine and **guanfacine** may decrease milk production.[76][81] Because it is transferred into breast milk, clonidine may cause side effects (e.g., hypotension) in the nursing infant.[81] Other options are preferred.

There is limited experience with **risperidone** during breastfeeding; other options are preferred.[81]

A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- When selecting a pharmacologic regimen, consider the child's daily schedule, predominant ADHD symptoms, likelihood of adherence, patient preferences (including medication cost) and risk of adverse effects.
- Crossover trials of stimulant versus placebo or other stimulants ("n of 1" trials) can be helpful in assessing the response to medication in a patient for whom therapeutic benefit is uncertain.
- Sustained-release or long-acting stimulant medications should not be crushed or chewed. In younger children who are unable to swallow certain medications, note that Adderall XR, Dexedrine spansules, and Vyvanse and Biphentin capsules can be opened and sprinkled on soft foods such as yogourt or applesauce, and Vyvanse can be mixed with water.
- When switching from stimulant therapy to atomoxetine or an antidepressant, a lower dose of the stimulant can be continued and tapered over approximately 3 weeks while the new drug takes effect.
- Patients with ADHD are at higher risk of impulsive behaviour and substance use.[8] Treating ADHD with stimulants can decrease the risk of substance abuse.[48][49][82]
- Approximately 50% of children and adolescents with tic disorders also have ADHD. In general, ADHD symptoms are more disabling than tics in patients with both. Although **stimulants** may exacerbate tics in some individuals, they can be safely used in the majority of patients with both disorders.[83] **Clonidine** or **guanfacine** alone or in combination with stimulants are also effective in improving both ADHD and tic symptoms in patients with both conditions.[83]

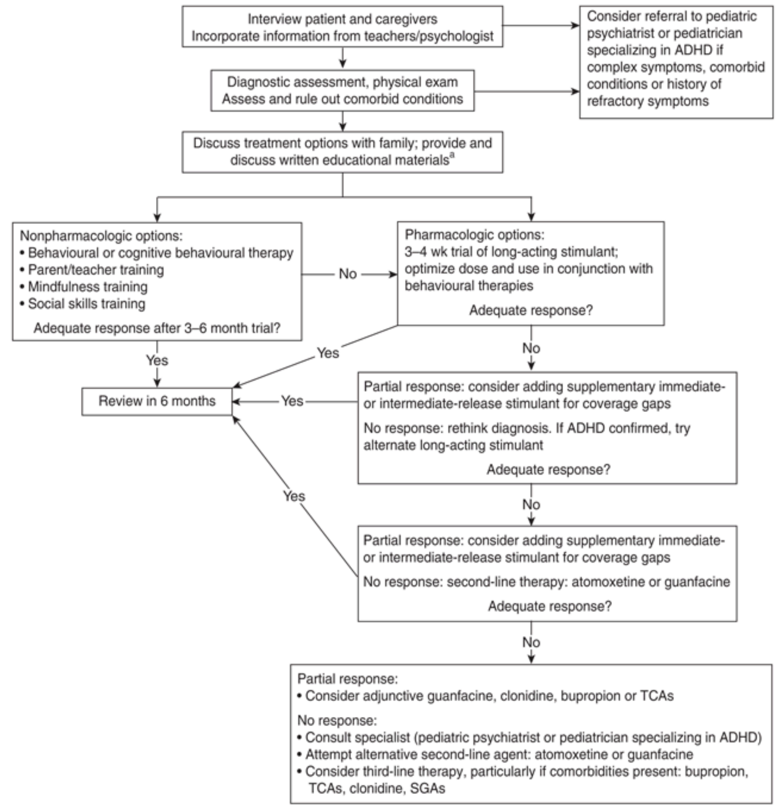
Resources

An ADHD medication chart is a useful resource for health-care professionals

Helpful information tips for teachers and parents are available from the U.S. Department of Education website.

Algorithms

Figure 1: *Management of Attention-Deficit Hyperactivity Disorder in Children*[8][28]



[a] See www.aboutkidshealth.ca.

Abbreviations: ADHD = attention-deficit hyperactivity disorder; SGA = second-generation antipsychotic; TCA = tricyclic antidepressant

Drug Tables

Table 3: Stimulants Used in the Management of Attention-Deficit Hyperactivity Disorder[84]

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
Drug Class: Amphetamine-based Agents					
dextroamphetamine immediate-release tablets Dexedrine, generics $30–60	2.5–5 mg BID (QAM and Q noon) PO; may increase by 2.5–5 mg Q 7 days Maximum: 6–17 y: 20–30 mg/day Adults: 50 mg/day	4 h	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics, e.g., chlorpromazine, fluphenazine. *Dextroamphetamine:* acidifying agents such as fruit juices or ascorbic acid can decrease absorption and increase elimination of dextroamphetamine. Alkalinizing agents such as sodium bicarbonate can increase absorption and decrease elimination of dextroamphetamine.	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar disorder and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months.
dextroamphetamine sustained-release spansules Dexedrine Spansule, generics $60–90	10 mg QAM PO; may increase by 2.5–5 mg Q 7 days Maximum: 6–17 y: 20–30 mg/day Adults: 50 mg/day	6–8 h (50% immediate-release, 50% delayed-release)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like"	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness.	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to risk prior to

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** “glassy eyes,” insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism.	Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics, e.g., chlorpromazine, fluphenazine.	prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar disorder and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Capsule contents can be sprinkled on soft food such as applesauce, ice cream or yogourt.
			If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		
lisdexamfetamine Vyvanse, generics $30–60	20–30 mg QAM PO; may increase by 10 mg Q 7 days Maximum: ≥6 y: 60 mg/day	13–14 h	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** “zombie-like” effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise.	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics, e.g., chlorpromazine, fluphenazine.	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar disorder and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Capsule contents can be mixed

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			Overdose symptoms—stop and retitrate: "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		with water or sprinkled on soft food such as applesauce, ice cream or yogourt. Available as a chewable tablet.
mixed salts amphetamine extended-release capsules Adderall XR, generics < $30	5–10 mg QAM PO; may increase by 5 mg Q 7 days Maximum: 6–12 y: 30 mg/day ≥13 y: 20–30 mg/day	12 h (50% immediate-release, 50% delayed-release)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain.	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential
			Transient—stop and re-evaluate: "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic	tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics, e.g., chlorpromazine, fluphenazine.	for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar disorder and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Capsule contents can be sprinkled on applesauce and eaten immediately without chewing; doses should not be divided or stored.

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		
Drug Class: Methylphenidate-based Agents					
methylphenidate immediate-release tablets generics < $30	5 mg BID–TID PO May increase by 5 mg Q 7 days Maximum: ≥6 y: 60 mg/day	3–4 h	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose based on response.
methylphenidate sustained-release tablets generics < $30	Once titrated to 20 mg/8 h of immediate-release product: 20 mg QAM PO	8 h	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability,	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
	Maximum: ≥6 y: 60 mg/day		dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular	hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		be predictive of clinical effect; titrate dose based on response. May be used in combination with immediate-release formulation.
methylphenidate controlled-release capsules Biphentin, generics $60–90	10–20 mg QAM PO; may increase by 5–10 mg Q 7 days Maximum: 6–17 y: 60 mg/day Adults: 80 mg/day Patients taking immediate-release formulations of methylphenidate can be converted to the next lower strength of Biphentin based on the total daily methylphenidate dose	10–12 h (40% immediate-release, 60% gradual effect)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias.	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase	Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose based on response. Capsule contents can be sprinkled on soft food such as applesauce, ice cream or yogourt.
methylphenidate controlled-release capsules	25 QAM PO; may increase by 10–15 mg Q 5 days	16 h (20% immediate-release, 80%	**Common, usually transient—continue**	*Stimulants:* avoid MAOIs, such as phenelzine,	Onset of effect usually seen in the first wk.

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
Foquest $120–150	Maximum: 6–17 y: 70 mg/day Adults: 100 mg/day Patients already taking methylphenidate can be converted to the next lower strength of Foquest based on the total methylphenidate daily dose	delayed controlled-release)	**therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of	tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose based on response. Capsule contents can be sprinkled on soft food, such as applesauce, ice cream or yogourt.
methylphenidate bilayer controlled-release tablets Concerta, generics $30–60	18 mg QAM PO; may increase by 9–18 mg Q 7 days Maximum: 6–17 y: 54 mg/day Adults: 72 mg/day Consult product monograph for dosage conversion from other methylphenidate formulations	12 h (22% immediate-release, 78% long-acting release)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** "zombie-like" effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	Generic product is bioequivalent, but clinical equivalence is unknown. CADDRA considers them nonequivalent due to earlier Tmax, shorter duration of effect, and some reports of patients destabilizing after switching.[8] Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect;

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
					titrate dose based on response.
methylphenidate delayed-release and extended-release capsules Jornay PM $120–150	20 mg QPM between 6:30–9:30 p.m. PO; may increase by 20 mg Q 7 days Maximum: 100 mg/day or 3.7 mg/kg/day Start dosing at 8 p.m. and adjust based on results. Range: 6:30–9:30 p.m. Dosage conversion from other methylphenidate products not recommended; discontinue and retitrate.	8–10 h delay, followed by a controlled release	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** “zombie-like” effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** “glassy eyes,” insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	Only formulation with evening administration; theoretically results in earlier symptom control next day. Capsule contents (beads) can be sprinkled on applesauce; beads should not be chewed. Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose based on response.
methylphenidate extended-release suspension Quillivant ER Powder for Oral Suspension $120–150	20 mg QAM PO; may increase by 10–20 mg Q 7 days Maximum: 60 mg/day Dosage conversion from other methylphenidate products not recommended;	12 h (20% immediate-release, 80% long-acting release)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:**	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations,	Oral suspension and chewable tablets are not interchangeable. Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
	discontinue and retitrate.		“zombie-like” effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise. **Overdose symptoms—stop and retitrate:** “glassy eyes,” insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism.	dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose
			If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]		based on response.
methylphenidate extended-release chewable tablet Quillivant ER Chewable Tablet $90–120	20 mg QAM PO; may increase by 10–20 mg Q 7 days Maximum: 60 mg/day Dosage conversion from other methylphenidate products not recommended; discontinue and retitrate.	8–12 h (30% immediate-release, 70% long-acting release)	**Common, usually transient—continue therapeutic trial:** anorexia, insomnia, weight loss, irritability, dizziness, weepiness, headache, abdominal pain. **Transient—stop and re-evaluate:** “zombie-like” effects, psychotic reactions (such as hallucinations), agitation, tachycardia, hypertension, growth failure, rebound hyperactivity, leukopenia, blood dyscrasias. Monitor patient for suicidal thoughts/ideation; consider a change in treatment if concerns arise.	*Stimulants:* avoid MAOIs, such as phenelzine, tranylcypromine and moclobemide, as they can increase hypertensive effect of stimulant. Theophylline may increase risk of tachycardia, palpitations, dizziness, weakness. Caution with SSRIs and SNRIs; risk of serotonin syndrome. Levels of stimulants may be altered by TCAs and antipsychotics (e.g., chlorpromazine, fluphenazine). Methylphenidate may increase plasma levels of phenytoin, phenobarbital, TCAs. Methylphenidate reduces metabolism	Oral suspension and chewable tablets are not interchangeable. Onset of effect usually seen in the first wk. Monitor BP and heart rate at baseline and within 1–3 months at follow-up. CNS stimulants are associated with the potential for abuse and dependence. Assess for this risk prior to prescribing and monitor for signs of abuse and dependence while on therapy. Monitor for bipolar and screen for manic symptoms. Monitor for growth in pediatric patients, including height and weight

Drug/Cost[a]	Dosage[b]	Duration of Action	Adverse Effects	Drug Interactions	Comments
			Overdose symptoms—stop and retitrate: "glassy eyes," insomnia, hyperactivity. **Significant:** sudden cardiac death reported; neurologic symptoms; exacerbation of tics; avoid in patients with a history of cardiovascular conduction disturbances, hypertension, acute psychotic episodes and hyperthyroidism. If seizures occur, or if frequency increases in patient with controlled epilepsy, stop and re-evaluate.[85]	of warfarin, resulting in increased INR. Carbamazepine decreases plasma levels of methylphenidate.	every 3–6 months. Last daily dose should be given before 4 p.m. to avoid insomnia. Doses greater than 60 mg/day usually do not result in additional efficacy in children. Pharmacokinetics vary widely among individuals; strict weight-based dosing may not be predictive of clinical effect; titrate dose based on response.

[a] Cost of 30-day supply of mean dosage; includes drug cost only.
[b] Dosages are based on CADDRA guidelines. Maximum doses may exceed manufacturer recommendations and are considered off-label use.[84]
Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.
Abbreviations: BP = blood pressure; CADDRA = Canadian ADHD Resource Alliance; CNS = central nervous system; INR = International Normalized Ratio; MAOI = monoamine oxidase inhibitor; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; TCA = tricyclic antidepressant

Table 4: *Other Agents Used in the Management of Attention-Deficit Hyperactivity Disorder*

Drug/Cost[a]	Dosage[b]	Adverse Effects	Drug Interactions	Comments
Drug Class: Alpha$_2$-adrenergic Agonists				
clonidine generics <$15	Initial: 0.05–0.1 mg/day PO Usual: 0.003–0.01 mg/kg/day PO (0.05–0.4 mg/day), once daily or in divided doses To discontinue, reduce dose by 0.1 mg/wk	Hypotension, sedation and dizziness common initially; dry mouth; could exacerbate depression.	Avoid concurrent use with TCAs. Additive effects with other CNS depressants, such as ethanol.	Typically improves hyperactivity and impulsiveness more than inattention. Onset of effect is usually 2–3 wk. Caution in patients with cardiovascular disease or depression.
guanfacine Intuniv XR, generics ~$80	1 mg daily PO; may increase by 1 mg Q 7–14 days Maximum: 6–12 y: 4 mg/day 13–17 y: 7 mg/day for monotherapy and 4 mg/day for adjunctive therapy To discontinue, reduce dose by 1 mg/wk	Somnolence, headache, fatigue, upper abdominal pain, irritability, emotional lability, nightmares, bradycardia, hypotension (minor).	Avoid concurrent use with TCAs. Additive effects with other CNS depressants, such as ethanol. Inhibitors of CYP3A4, such as clarithromycin and ketoconazole, will significantly increase guanfacine serum concentrations. Inducers of CYP3A4, such as carbamazepine, systemic dexamethasone, phenobarbital, phenytoin and rifampin, will significantly reduce guanfacine serum concentrations.	Onset of effect is usually after 2 wk. Selective alpha$_{2a}$-adrenergic agonist; less sedation and hypotension than clonidine. In Canada, indicated only in children 6–17 y. During and after discontinuation, monitor BP and heart rate until they return to normal.

Drug/Cost[a]	Dosage[b]	Adverse Effects	Drug Interactions	Comments
			Caution with agents that decrease heart rate and/or BP (beta blockers) and QTc prolonging agents (quetiapine, quinidine).	
Drug Class: Antidepressants, dual-action				
bupropion Wellbutrin SR, Odan–Bupropion SR, other generics <$15	Initial: 2–3 mg/kg/day PO Usual: 200–300 mg/day PO in 2 divided doses Single doses should not exceed 150 mg	Agitation, dry mouth, insomnia, headache, constipation, nausea, vomiting, nervousness, dizziness, sweating, hypertension and tachycardia, suicidal ideation, seizures (0.5–1% incidence). Avoid in patients with a history of seizure disorders, eating disorders or significant head injury. Seizure risk increased with doses >300 mg/day.	Inducers of CYP2D6, CYP2B6 or CYP3A4, such as carbamazepine, phenytoin or rifampin, may decrease plasma level of bupropion and increase level of hydroxybupropion (active metabolite). Plasma concentration of venlafaxine or TCAs, such as imipramine, desipramine or nortriptyline, may increase due to inhibition of CYP2D6 by bupropion. Avoid use with MAOIs; may cause mania, excitation, hyperpyrexia. Be vigilant for initiation of other medications that can decrease the seizure threshold, such as chloroquine, theophylline, tramadol or ciprofloxacin.	May take 2–4 wk before effects on ADHD symptoms are seen.
Drug Class: Antidepressants, serotonin-norepinephrine reuptake inhibitors (SNRIs)				
venlafaxine Effexor XR, generics <$15	Adults: Initial: 37.5–75 mg daily PO for 1 wk; titrate gradually to 150–300 mg daily PO Maximum: 375 mg/day PO	Nausea, drowsiness, nervousness, dizziness, dry mouth, may increase BP if dose >300 mg/day.	Inhibitors of CYP2D6 or CYP3A4, such as clarithromycin, erythromycin, grapefruit, fluoxetine or paroxetine, may increase venlafaxine levels; avoid use with MAOIs; caution with other serotonergic drugs (serotonin syndrome).	May take up to 4 wk for optimal drug effect.
Drug Class: Antidepressants, tricyclic				
desipramine generics <$15–30	6–12 y: 10–20 mg/day PO in 3–4 divided doses Adolescents: 30–50 mg/day PO in 3–4 divided doses Usual maximum: 150 mg/day	Postural hypotension, anticholinergic effects (dry mouth, constipation, urinary retention), dizziness, nausea, drowsiness, weakness, tremor, weight gain, asymptomatic ECG changes, tachycardia and arrhythmias (potential for fatal arrhythmias in overdose). Where possible, avoid in patients with a history of cardiovascular conduction disturbances, urinary retention, seizure disorders, hyperthyroidism.	Avoid with MAOIs; may cause mania, excitation, hyperpyrexia. Inducers of CYP2D6 or CYP3A4, such as carbamazepine, phenytoin or rifampin, may decrease effect; inhibitors of these isoenzymes, such as clarithromycin, erythromycin, grapefruit juice, fluoxetine or paroxetine, may increase effect and toxicity.	May take 3–4 wk to see beneficial effects.
imipramine generics <$15	6–12 y: 10–20 mg/day PO in 3–4 divided doses Adolescents: 30–50 mg/day PO in 3–4	Postural hypotension, anticholinergic effects (dry mouth, constipation, urinary retention), dizziness, nausea, drowsiness,	Avoid with MAOIs; may cause mania, excitation, hyperpyrexia. Inducers of CYP2D6 or CYP3A4, such as carbamazepine,	May take 3–4 wk to see beneficial effects.

Drug/Cost[a]	Dosage[b]	Adverse Effects	Drug Interactions	Comments
	divided doses Usual maximum: 150 mg/day	weakness, tremor, weight gain, asymptomatic ECG changes, tachycardia and arrhythmias (potential for fatal arrhythmias in overdose). Where possible, avoid in patients with a history of cardiovascular conduction disturbances, urinary retention, seizure disorders, hyperthyroidism.	phenytoin or rifampin, may decrease effect; inhibitors of these isoenzymes, such as clarithromycin, erythromycin, grapefruit juice, fluoxetine or paroxetine, may increase effect and toxicity.	
nortriptyline Aventyl <$15–30	6–12 y: 10–20 mg/day PO in 3–4 divided doses Adolescents: 30–50 mg/day PO in 3–4 divided doses Usual maximum: 150 mg/day	Postural hypotension, anticholinergic effects (dry mouth, constipation, urinary retention), dizziness, nausea, drowsiness, weakness, tremor, weight gain, asymptomatic ECG changes, tachycardia and arrhythmias (potential for fatal arrhythmias in overdose). Where possible, avoid in patients with a history of cardiovascular conduction disturbances, urinary retention, seizure disorders, hyperthyroidism.	Avoid with MAOIs; may cause mania, excitation, hyperpyrexia. Inducers of CYP2D6 or CYP3A4, such as carbamazepine, phenytoin or rifampin, may decrease effect; inhibitors of these isoenzymes, such as clarithromycin, erythromycin, grapefruit juice, fluoxetine or paroxetine, may increase effect and toxicity.	May take 3–4 wk to see beneficial effects.
Drug Class: Antipsychotics, second-generation				
risperidone Risperdal, other generics <$15	Initial: 0.25–0.5 mg HS PO; increase at weekly intervals by 0.5 mg/day PO in 2 divided doses as needed Usual maintenance dose: 0.75–1.5 mg/day PO	Weight gain, drowsiness, headache, orthostatic hypotension, dyspepsia, dose-related extrapyramidal effects; hyperprolactinemia. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.	Inhibitors of CYP2D6 or CYP3A4, such as clarithromycin, erythromycin, grapefruit, fluoxetine or paroxetine, may increase effect and toxicity; inducers of these isoenzymes, such as carbamazepine, phenytoin or rifampin, may reduce effect.	Used for behavioural symptoms (hyperactivity, impulsivity) when stimulants alone are ineffective or not tolerated. Oral liquid should not be mixed with cola or tea; may be mixed with water, orange juice or low-fat milk.
Drug Class: Norepinephrine Reuptake Inhibitors				
atomoxetine generics $15–30	6–17 y and <70 kg: 0.5 mg/kg/day PO × 10 days, then 0.8 mg/kg/day PO × 10 days, then 1.2 mg/kg/day PO 6–17 y and >70 kg and adults: 40 mg/day PO × 10 days, then 60 mg/day PO × 10 days, then increase to target of 80 mg/day PO if necessary Maximum: ≥6 y: 1.4 mg/kg/day, not to exceed 100 mg/day	Headache, rhinorrhea, upper abdominal pain, nausea, sedation, vomiting, decreased appetite, dizziness, fatigue, emotional lability, and small increases in heart rate and BP. Significant: suicidal ideation, sudden cardiac death, liver toxicity. May increase QTc interval.	Inhibitors of CYP2D6, such as fluoxetine, paroxetine or quinidine, can increase plasma levels. Concurrent use of salbutamol may increase heart rate. Contraindicated with MAOIs.	Requires 3–4 wk to see beneficial effects. In normal or rapid metabolizers of CYP2D6, dosing to 1.2 mg/kg/day may be titrated more rapidly. Slow metabolizers require a slower titration depending on their isoenzyme phenotype.[86]

[a] Cost of 30-day supply of mean dosage; based on 35 kg body weight; includes drug cost only.
[b] Dosages are based on CADDRA guidelines where applicable. Maximum doses may exceed manufacturer recommendations and are considered off-label use.[84]
Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.

Abbreviations: ADHD = attention-deficit hyperactivity disorder; BP = blood pressure; CNS = central nervous system; ECG = electrocardiogram; MAOI = monoamine oxidase inhibitor; TCA = tricyclic antidepressant

Suggested Readings

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